



Department of Computer Engineering

Academic Year: 2025-26

Semester: IV

Subject: DBMS Lab

Year: SE

Name of Student:

Roll No.:

Moodle Id:

Date of Performance:

Date of Submission:

Name of Instructor: Prof. Archana Kotangale

Experiment No.

Aim: : Implement DML commands for a given database system. Perform data insertion, update, and deletion operations, and write simple SQL queries using string manipulation and aggregate functions to retrieve meaningful information.

Program:

```
mysql> select * from clients;
```

ClientNo	Name	city	PinCode	State	BalDue	Amount	DateOP
C01	Ivan Bayross	Mumbai	400054	Maharashtra	15000	25000	2019-02-03
C02	Mamta Muzumdar	Madras	780001	Tamil Nadu	10000	30000	2019-12-04
C03	Ritika Nachare	Mumbai	400057	Maharashtra	10000	35000	2019-12-30
C04	Pranav Kumar	Banglore	560001	Karnataka	1500	50000	2019-06-30
C05	Krishna prasad	Banglore	400060	Maharashtra	2000	10000	2019-03-31
C06	Lavanya Revankar	Manglore	560050	Karnataka	6000	22000	2019-07-02

```
6 rows in set (0.00 sec)
```

Output:



1. Listing of the names of all clients having 'a' as the second letter in their names

```
mysql> select Name
-> from clients
-> where Name like '_a%';
+-----+
| Name          |
+-----+
| Mamta Muzumdar |
| Lavanya Revankar |
+-----+
```

2. Listing of the clients who stay in a city whose first letter is 'M'.

```
mysql> select * from clients where city like 'M%';
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name          | city    | PinCode | State      | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C01      | Ivan Bayross  | Mumbai  | 400054  | Maharashtra | 15000  | 25000  | 2019-02-03 |
| C02      | Mamta Muzumdar | Madras  | 780001  | Tamil Nadu  | 10000  | 30000  | 2019-12-04 |
| C03      | Ritika Nachare | Mumbai  | 400057  | Maharashtra | 10000  | 35000  | 2019-12-30 |
| C06      | Lavanya Revankar | Manglore | 560050  | Karnataka  | 6000   | 22000  | 2019-07-02 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

3. Display Name city and balance using concatenation function, as Ivan Byross Mumbai having balance 15000

```
mysql> select concat(Name, " ", city, " having balance ", BalDue) from clients where ClientNo='C01';
+-----+
| concat(Name, " ", city, " having balance ", BalDue) |
+-----+
| Ivan Bayross Mumbai having balance 15000           |
+-----+
1 row in set (0.02 sec)
```



4. Display names in uppercase

```
mysql> SELECT UPPER(Name) FROM clients;
+-----+
| UPPER(Name) |
+-----+
| IVAN BAYROSS |
| MAMTA MUZUMDAR |
| RITIKA NACHARE |
| PRANAV KUMAR |
| KRISHNA PRASAD |
| LAVANYA REVANKAR |
+-----+
6 rows in set (0.00 sec)
```

5. Find the unique states present in the client table.

```
mysql> SELECT DISTINCT State FROM clients;
+-----+
| State |
+-----+
| Maharashtra |
| Tamil Nadu |
| Karnataka |
+-----+
```

6. Display records by ascending order of date of purchase

```
mysql> SELECT * FROM clients ORDER BY DateOP ASC;
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name | city | PinCode | State | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+
| C01 | Ivan Bayross | Mumbai | 400054 | Maharashtra | 15000 | 25000 | 2019-02-03 |
| C05 | Krishna prasad | Banglore | 400060 | Maharashtra | 2000 | 10000 | 2019-03-31 |
| C04 | Pranav Kumar | Banglore | 560001 | Karnataka | 1500 | 50000 | 2019-06-30 |
| C06 | Lavanya Revankar | Manglore | 560050 | Karnataka | 6000 | 22000 | 2019-07-02 |
| C02 | Mamta Muzumdar | Madras | 780001 | Tamil Nadu | 10000 | 30000 | 2019-12-04 |
| C03 | Ritika Nachare | Mumbai | 400057 | Maharashtra | 10000 | 35000 | 2019-12-30 |
+-----+-----+-----+-----+-----+-----+-----+
6 rows in set (0.01 sec)
```



7. List all clients who stay in 'Bangalore' or 'Mangalore'.

```
mysql> select * from clients
-> where city='Bangalore' or city='Mangalore';
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name           | city      | PinCode | State      | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C04      | Pranav Kumar   | Bangalore | 560001  | Karnataka  | 1500   | 50000  | 2019-06-30 |
| C05      | Krishna prasad | Bangalore | 400060  | Maharashtra | 2000   | 10000  | 2019-03-31 |
| C06      | Lavanya Revankar | Manglore | 560050  | Karnataka  | 6000   | 22000  | 2019-07-02 |
+-----+-----+-----+-----+-----+-----+-----+-----+
3 rows in set (0.04 sec)
```

8. List all clients whose BalDue is greater than value 10,000

```
mysql> select * from clients where BalDue>10000;
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name           | city      | PinCode | State      | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C01      | Ivan Bayross   | Mumbai    | 400054  | Maharashtra | 15000  | 25000  | 2019-02-03 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

9. Listing of names, city, and state of clients who are not in the state of 'Maharashtra'.

```
mysql> SELECT Name, city, State FROM clients WHERE State != 'Maharashtra';
+-----+-----+-----+
| Name           | city      | State      |
+-----+-----+-----+
| Mamta Muzumdar | Madras    | Tamil Nadu |
| Pranav Kumar   | Bangalore | Karnataka  |
| Lavanya Revankar | Manglore | Karnataka  |
+-----+-----+-----+
3 rows in set (0.00 sec)
```



10. Find the average BalDue of the clients.

```
mysql> SELECT AVG(BalDue) AS average_balance_due FROM clients;
+-----+
| average_balance_due |
+-----+
|          7416.6667 |
+-----+
```

11. Find minimum and maximum BalDue among the client

```
mysql> SELECT MIN(BalDue) FROM clients;
\+-----+
| MIN(BalDue) |
+-----+
|          1500 |
+-----+
1 row in set (0.02 sec)

mysql> SELECT MAX(BalDue) FROM clients;
+-----+
| MAX(BalDue) |
+-----+
|          15000 |
+-----+
1 row in set (0.00 sec)
```

12. Find the name of the clients whose BalDue is between 1000 to 2000.

```
mysql> select * from clients where BalDue>1000 and BalDue<2000;
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name          | city      | PinCode | State      | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C04      | Pranav Kumar | Bangalore | 560001  | Karnataka | 1500   | 50000  | 2019-06-30 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```



13. Find total of the BalDue as 'TotalDueBal' from given table

```
mysql> SELECT SUM(BalDue) AS TotalBalDue FROM clients;
+-----+
| TotalBalDue |
+-----+
|         44500 |
+-----+
1 row in set (0.00 sec)
```

14. Find the clients who belongs to Karnataka.

```
mysql> SELECT * FROM clients WHERE State = 'Karnataka';
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name           | city      | PinCode | State      | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C04      | Pranav Kumar   | Banglore  | 560001  | Karnataka  | 1500   | 50000  | 2019-06-30 |
| C06      | Lavanya Revankar | Manglore  | 560050  | Karnataka  | 6000   | 22000  | 2019-07-02 |
+-----+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

15. Count the number of clients from Maharashtra

```
mysql> SELECT COUNT(ClientNo) FROM clients where State='Maharashtra';
+-----+
| COUNT(ClientNo) |
+-----+
|                3 |
+-----+
1 row in set (0.00 sec)
```



16. Find the clients who live in Maharashtra and whose balance due is greater than 10000.

```
mysql> SELECT * FROM clients where State='Maharashtra' and BalDue>10000;
+-----+-----+-----+-----+-----+-----+-----+-----+
| ClientNo | Name       | city   | PinCode | State       | BalDue | Amount | DateOP |
+-----+-----+-----+-----+-----+-----+-----+-----+
| C01      | Ivan Bayross | Mumbai | 400054 | Maharashtra | 15000  | 25000  | 2019-02-03 |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

17. Update the city of the client C04 to Pune, state to Maharashtra and Pincode 400019.

```
mysql> UPDATE clients SET city = 'Pune',
Query OK, 1 row affected (0.08 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> 
```

18. Delete the record of C06 from database

```
mysql> DELETE FROM clients WHERE ClientNo = 'C06';
Query OK, 1 row affected (0.04 sec)

mysql> 
```



Exercise 2: Apply various string manipulation functions (such as CONCAT, UPPER, LOWER, SUBSTRING, TRIM) on the data in the table created in Experiment 3.

Concat

```
mysql> SELECT CONCAT(first_name, ' ', last_name) AS full_name FROM employees;
ERROR 1146 (42S02): Table 'election.employees' doesn't exist
mysql> SELECT CONCAT(start_date, ' ', end_date) AS election_dates FROM elections
;
+-----+
| election_dates |
+-----+
| 2025-12-12 2025-11-12 |
| 2025-12-12 2025-11-12 |
| 2025-12-12 2025-11-12 |
+-----+
3 rows in set (0.00 sec)
```

Upper

```
mysql> UPDATE candidate SET fname = UPPER(fname);
Query OK, 0 rows affected (0.00 sec)
Rows matched: 2 Changed: 0 Warnings: 0
```

```
mysql> select * from candidate;
+-----+-----+-----+-----+-----+-----+-----+
| candidate_id | eid | caste | gender | fname | lname | area |
+-----+-----+-----+-----+-----+-----+-----+
| 101 | 221 | Hindu | F | MEENA | Moy | Thane |
| 102 | 222 | Marathi | M | ROHIT | Ken | Mumbai |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```



Lower

```
mysql> UPDATE candidate SET fname = Lower(lname);
Query OK, 2 rows affected (0.04 sec)
Rows matched: 2  Changed: 2  Warnings: 0

mysql> select * from candidate;
+-----+-----+-----+-----+-----+-----+-----+
| candidate_id | eid | caste | gender | fname | lname | area |
+-----+-----+-----+-----+-----+-----+-----+
|          101 | 221 | Hindu | F      | moy   | Moy   | Thane |
|          102 | 222 | Marathi | M      | ken   | Ken   | Mumbai |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

Substring

```
mysql> SELECT SUBSTRING('SQL Tutorial', 5);
+-----+
| SUBSTRING('SQL Tutorial', 5) |
+-----+
| Tutorial                      |
+-----+
1 row in set (0.00 sec)

mysql> -- Result: 'utorial'
```

Trim

```
mysql> SELECT TRIM(LEADING '0' FROM '00092500');
+-----+
| TRIM(LEADING '0' FROM '00092500') |
+-----+
| 92500                               |
+-----+
1 row in set (0.00 sec)
```

Conclusion: DML commands in SQL are essential for managing and manipulating data within tables. They allow users to insert, update, delete, and retrieve records efficiently. DML also supports string manipulation functions to modify or query text data effectively. Proper use of these commands ensures accurate, organized, and flexible data handling in a database.